Quantitative Research & Trading Framework
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Statistical Arbitrage via PCA-Residual Mean Reversion

A Detailed Literature Review

on Indian Equities, with an ML and Intraday Extension

Phase 2 Deliverable — Topic Selection & Detailed Literature Review
Selected strategy family: *market-neutral PCA-residual statistical arbitrage*
Framing: *empirical hypothesis test, not a guaranteed-return system*

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Executive Summary

This review supports the selection of **PCA-residual statistical arbitrage** as the strategy family for the project. The approach decomposes equity returns into common (systematic) factors and idiosyncratic residuals via Principal Component Analysis, models each residual as a mean-reverting Ornstein–Uhlenbeck process, and trades the standardised deviation (the s-score) market-neutrally. The canonical reference is Avellaneda and Lee (2010), who construct, back-test and compare PCA- and ETF-based statistical arbitrage on the broad US equity universe.

The review establishes four things. First, the strategy is a genuine, published, professional paradigm with a clear mechanism. Second, the reported edge is *modest and has decayed over time*: Avellaneda and Lee report an average Sharpe ratio of roughly 0.9 for PCA-based strategies over 2003–2007, rising to 1.51 for ETF-based strategies augmented with trading-volume information — figures that already reflect a documented degradation since the early 2000s. Third, the strategy has well-understood failure modes: the factor structure breaks down during regime shifts and crowded de-leveraging events (the August 2007 "Quant Quake" analysed by Khandani and Lo), and transaction costs threaten the small per-trade edge. Fourth, and most importantly for this project, the right posture is to treat profitability on Indian equities as an *empirical hypothesis to be tested honestly*, with proper out-of-sample evaluation and the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) to guard against backtest overfitting. A clean negative result, properly analysed, is a valid and valuable outcome consistent with the project brief.

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1. Problem Statement and Market Setting

1.1 The research question

The project brief frames the core question conditionally: *under what market conditions, on what instruments, and at what time horizons do which strategies become useful?* This review narrows that question to a single, well-defined strategy family and asks: *does PCA-residual mean reversion, applied to liquid Indian equities and subjected to realistic transaction costs, retain a usable edge — and if so, under which regimes?*

1.2 Market setting

The intended market is the liquid segment of the Indian equity cash market (NSE), specifically a universe of roughly 250–300 names filtered from the Nifty 500 for liquidity, price floor, and trading history. The relevant horizon is short-to-medium: holding periods from intraday up to approximately two to three weeks, which is the natural timescale over which idiosyncratic residuals revert. This horizon places the strategy squarely in the *mid-frequency* band occupied by quantitative statistical-arbitrage desks, between high-frequency market making and slow factor investing.

1.3 Why this matters

A strategy is not merely predictive; per the brief, it "must also survive transaction costs . . . slippage, and changing market conditions." Statistical arbitrage is an especially demanding test of this principle because it is high-turnover by construction: the edge per trade is small, and costs are paid repeatedly. The Indian cost environment — in particular the Securities Transaction Tax (STT) — makes the cost constraint binding rather than incidental. This review therefore treats cost realism and honest evaluation as first-class concerns, not afterthoughts.

2. The Strategy Family: From Pairs to PCA Residuals

2.1 Relative-value mean reversion: the lineage

The strategy belongs to the relative-value mean-reversion family, whose canonical empirical study is the pairs-trading work of Gatev, Goetzmann, and Rouwenhorst (2006). They test the classic Wall-Street "pairs trading" rule on US daily data over 1962–2002: stocks are matched into pairs by minimum distance between normalised price series, and positions are opened when the spread diverges and closed when it converges. They report average annualised excess returns of up to roughly 11% for self-financing portfolios of pairs, exceeding conservative

transaction-cost estimates, and—crucially—attribute the profitability to "temporary mis-pricing of close substitutes" linked to "the presence of a common factor in the returns." Two lessons carry directly into the present design: (i) the source of profit is temporary, non-fundamental dislocation; and (ii) the relationship being traded is governed by a common factor.

2.2 The PCA generalisation

PCA-residual statistical arbitrage generalises pairs trading from two hand-matched stocks to the entire universe at once. Rather than choosing substitutes by hand, one estimates the common factors *statistically* via PCA and trades the residual that remains after those factors are removed. Avellaneda and Lee (2010) formalise this. They model stock returns with a multi-factor system,

$$\frac{dS_i(t)}{S_i(t)} = \alpha_i\,dt + \sum_{j=1}^{m} \beta_{ij}\,F_j(t) + dX_i(t), \tag{1}$$

where the F_j are common factors (estimated by PCA on the correlation matrix of returns, or alternatively proxied by sector ETFs), β_{ij} are factor loadings ("exposures"), and $X_i(t)$ is the idiosyncratic residual. The residual is modelled as a mean-reverting Ornstein–Uhlenbeck process,

$$dX_i(t) = \kappa_i\big(\mu_i - X_i(t)\big)\,dt + \sigma_i\,dW_i(t), \qquad \kappa_i > 0, \tag{2}$$

whose mean-reversion speed κ_i determines a characteristic reversion time $\tau_i = 1/\kappa_i$. The standardised deviation of the residual from its equilibrium defines the dimensionless trading signal, the s-score,

$$s\text{-score}_i = \frac{X_i(t) - \mu_i}{\sigma_{eq,i}}, \tag{3}$$

and "contrarian" rules follow naturally: short when the s-score is large and positive (the stock is rich relative to its factor-implied value), long when it is large and negative, and exit as the s-score reverts toward zero.

> **Key mechanism.** PCA does not predict; it *decomposes*. The same regression of a stock's returns on the estimated factors yields both the exposures (the loadings β_{ij}) and the residual X_i. Profit is hypothesised to come from the mean reversion of that residual — a modelling assumption, validated empirically, not a theorem.

2.3 Reported performance — and its decay

It is essential to be precise about what Avellaneda and Lee actually report, because it calibrates realistic expectations. Over their sample, PCA-based strategies achieved an average Sharpe ratio of approximately 0.9 (2003–2007); ETF-based strategies achieved approximately 1.1 (1997–2007); and ETF-based strategies augmented with trading-volume information achieved approximately 1.51 (2003–2007). The authors explicitly document degradation of these returns over time and relate strategy performance to the market cycle, studying in detail the behaviour during the liquidity crisis of summer 2007. The headline, therefore, is not a spectacular edge but a *modest, decaying* one — which is exactly why disciplined cost modelling and evaluation dominate the design.

3. Signal Construction: Estimation Choices and Dependencies

3.1 Estimating the common factors

PCA is applied to a rolling window of daily returns (Avellaneda and Lee use a window on the order of one year for factor estimation; shorter windows are common in faster variants). The

top several principal components are retained as the common factors. The leading component typically corresponds to a broad market mode; subsequent components capture sector and style structure. The number of retained components is a design parameter that trades off completeness of factor removal against estimation stability.

3.2 Estimating mean reversion and filtering

For each stock, the residual process is estimated and its mean-reversion speed κ_i recovered (in discrete time, by regressing the residual on its own lag; the slope b implies $\kappa = -\ln b$ per unit time). Avellaneda and Lee filter out names whose estimated reversion is too slow to be tradable on the intended horizon: an extreme s-score on a non-reverting residual is more likely to reflect genuine, unmodelled information than a temporary dislocation, and should not be traded. This "trade only what actually reverts" filter is a central robustness device.

3.3 Dependencies between estimation choices

The brief stresses mapping dependencies between factors. Three couple tightly here. (i) *Window length and horizon:* the factor-estimation window defines what counts as "normal," and must be long enough for stable factors yet short enough to remain relevant to a days-to-weeks horizon. (ii) *Number of factors and residual definition:* retaining more components removes more common variation, changing the residual that is traded. (iii) *Threshold and turnover:* the entry threshold on the s-score directly controls trade frequency and therefore cost drag — a higher threshold yields fewer, higher-conviction trades, a deliberate response to a binding cost constraint. None of these should be tuned to maximise a backtest; each should be set by reasoning and confirmed for non-fragility, for reasons made precise in Section 6.

4. Risk, Crowding, and Regime Dependence

4.1 The factor structure breaks in stress

Mean reversion assumes that divergences are temporary. This assumption fails precisely when divergences are *real* — during regime shifts, when correlations spike and the previously stable factor structure breaks down. In such periods the residual that the model treats as noise is in fact signal (a company genuinely re-rating), and the contrarian bet loses on both legs. Regime awareness is therefore not optional; it is structural to the strategy's survival.

4.2 Crowding and the 2007 Quant Quake

The most important risk specific to this strategy class is documented by Khandani and Lo (2007) in their analysis of the August 2007 "Quant Quake." During the week of 6 August 2007, quantitative long/short equity market-neutral funds — many running closely related statistical-arbitrage strategies — suffered unprecedented, simultaneous losses. Khandani and Lo hypothesise that a rapid unwind of one or more sizable market-neutral portfolios (likely a forced liquidation) moved prices against the very positions that similar funds collectively held, triggering further stop-loss and de-leveraging cascades, with a sharp rebound on 10 August consistent with the unwind hypothesis. The lesson is sobering and direct: positions that appear diversified across many names can behave as *one crowded bet* during stress, because many participants hold the same relative-value exposures. Diversification across names does not protect against this; only conservative sizing, regime-based de-risking, and hard draw-down limits soften it — and none eliminate it.

4.3 Implications for risk design

Three defences follow directly from the literature. (i) A *volatility-based regime throttle* (e.g. scaling exposure down as a volatility index rises, halting in crisis) addresses the regime-shift failure mode at low overfitting risk. (ii) *Hard, non-discretionary limits* — per-name caps, gross-exposure caps, and a draw-down circuit breaker — bound the damage from crowded unwinds. (iii) *Dollar-neutral* (and optionally sector-neutral) construction removes directional and sector bets so that only idiosyncratic reversion is traded. These are protective, not return-generating; their purpose is survival.

5. Machine Learning: Where It Adds Value (and Where It Does Not)

5.1 Principle

The ML literature for finance is candid about its difficulty: low signal-to-noise, non-stationarity, and the ease of overfitting mean that naive price prediction generally fails. The disciplined use, following López de Prado's *Advances in Financial Machine Learning* (2018), is to apply ML where a genuine pattern-recognition problem exists, and to leave well-functioning rules and hard safety constraints alone. For this strategy, that principle localises ML to two places — sizing and exit timing — and explicitly excludes two others — the signal itself (which is mathematical) and the risk limits (which must remain hard).

5.2 Meta-labelling for position sizing

The primary ML addition is *meta-labelling*. A primary rule (the s-score crossing its threshold) determines trade direction; a secondary model then predicts the probability that a given signal will succeed, and that probability scales the position. The model is trained on historical signals using the triple-barrier method (profit barrier, stop barrier, time barrier) to label outcomes. Two refinements from the source material are important. First, binary win/loss labels discard magnitude information, so training samples are *weighted by the magnitude of their realised P&L*, ensuring that large winners and losers exert proportionate influence. Second, because labels constructed from overlapping holding periods are not independent, samples are additionally weighted by *uniqueness* to mitigate the resulting leakage. A gradient-boosting model on tabular features (the s-score, reversion speed, residual age, volatility, volume, factor-fit quality, and market-regime indicators) is the natural default.

5.3 An ML exit model

A second, optional addition refines the exit decision: given the current state of an open trade (s-score, time held, profit net of costs, regime), a model decides whether to exit early or hold, trained on what historical price paths reveal to have been the better action. A single guardrail is non-negotiable and follows directly from the risk discussion: the hard stop-loss always remains active, so the model may exit early but can never hold a runaway loser. This guards against the disposition effect — "cutting winners and holding losers" — which produces small gains and catastrophic losses and is especially dangerous for a strategy whose tail risk is the non-reverting divergence.

5.4 What ML must not do

Two exclusions are deliberate. ML is not applied to the signal: "ML-ing" the s-score amounts to direct price prediction, the failure mode above. And ML is not applied to the risk limits: stops, caps, and the draw-down breaker must be hard and predictable. In short, ML refines *sizing* and *timing*; it never sets direction or overrides safety.

6. Evaluation Methodology

6.1 Why naive backtesting misleads

The central hazard is backtest overfitting through multiple testing: when many parameter combinations are tried, the best-looking Sharpe ratio is inflated by selection, even when no real edge exists. Bailey and López de Prado (2014) formalise this with the **Deflated Sharpe Ratio (DSR)**, which adjusts an observed Sharpe ratio for (i) the number of trials conducted, (ii) the sample length, and (iii) the skewness and kurtosis of returns, and reports the probability that the true Sharpe ratio is positive. The DSR makes honest trial-counting a methodological requirement: every variation tried must be recorded, or the correction is meaningless.

6.2 Protocol

The evaluation protocol combines several standard devices. *Walk-forward analysis* trains on a past window and tests on the subsequent window, rolling forward through the full history, so that every test observation is genuinely out-of-sample and many regimes are traversed. Reported metrics are a *panel*, not a single number: net (after-cost) Sharpe and Sortino ratios; maximum draw-down and draw-down duration; win rate together with the ratio of average win to average loss (to confirm losers are cut and winners are let run); and turnover (which links the strategy to its cost drag). The DSR provides the overfitting correction, and a *regime breakdown* (calm, crisis, choppy) tests robustness and validates that the regime filter actually protects the strategy — for example, by examining performance through the COVID-19 crash. A single, final out-of-sample read on held-back data, taken once and not re-optimised, provides the most honest performance estimate.

6.3 Transaction-cost model

Because costs are the binding constraint, they are modelled pessimistically. The dominant Indian-specific component is the Securities Transaction Tax; brokerage, the bid–ask spread, market impact, and exchange and regulatory charges contribute the remainder. A realistic round-trip cost on liquid names is on the order of 0.35–0.50%, against a gross per-trade edge plausibly in the range 0.3–0.8% averaged over winners and losers. The implied net edge is small and may be negative — which is the central empirical question, not a detail. The intraday variant exploits the lower intraday STT rate but constrains holding periods to a single session; whether intraday reversions are strong and frequent enough to overcome that constraint is itself a question worth testing.

7. Comparison Matrix

The table below situates the selected strategy against the families considered, on the dimensions the brief emphasises.

Strategy	Horizon	Data	Overfit risk	Practical viability (India)
PCA-residual stat-arb *(selected)*	Hours–weeks	Daily OHLCV (free); minute bars for intraday	Medium	Strong fit to horizon and skills; cost-constrained; free baseline data

Strategy	Horizon	Data	Overfit risk	Practical viability (India)
Pairs trading	Days–weeks	Daily OHLCV (free)	Low–medium	Good as a baseline; less scalable than PCA
Time-series momentum	Weeks–months	Daily futures/equity	Lowest	Robust but slower than target horizon
Cross-sectional factors	Monthly	Prices + fundamentals	Low–medium	Slower to iterate; fundamentals harder to source cleanly
Order-book imbalance	Seconds–minutes	LOB / tick (costly)	High	Infrastructure- and data-prohibitive; not pursued

The selected family is the best joint match to the target horizon (intraday to a few weeks), the available data (free daily data suffices for the baseline), the relevant skill set (PCA, optimisation, portfolio construction), and the brief's emphasis on conditional, regime-dependent edge with rich, studyable failure modes.

8. Conclusion and Recommended Direction

8.1 Selection

The recommended direction is **market-neutral PCA-residual statistical arbitrage on liquid Indian equities**, built classical-first and then refined: (1) a pure-mathematical baseline (PCA $\rightarrow$ residual $\rightarrow$ s-score) with fixed entry/exit rules, dollar-neutral construction, a volatility regime filter, and hard risk limits; (2) an ML meta-labelling layer for position sizing; (3) an optional ML exit model, with the hard stop-loss always retained; and (4) an intraday live-price extension as a deliberate, later progression. This ordering follows the brief's prescribed build sequence and ensures each added layer is measured against the one below it.

8.2 Honest expectations

The literature supports neither dismissal nor over-optimism. The mechanism is real and professionally used; the reported edge is modest (PCA Sharpe near 0.9 in the source study) and has decayed; the failure modes (regime breakdown, crowded unwinds, costs) are well documented; and the Indian cost environment makes profitability genuinely uncertain. Accordingly, the project's contribution is framed as a *rigorous, cost-honest hypothesis test*: to establish whether, and under which regimes, PCA-residual mean reversion retains a usable edge on Indian equities. Consistent with the brief's explicit guidance that "negative results are valuable if analysed properly," a clean negative result — demonstrated with proper out-of-sample evaluation and a Deflated Sharpe Ratio that honestly accounts for every variation tried — is a legitimate and informative outcome.

8.3 Open items deferred by design

Two extensions are noted but deliberately left unspecified at this stage: a multi-signal combination layer (relevant only once more than one signal exists) and a news-sentiment / agentic layer (e.g. GDELT news with a FinBERT sentiment model). These are recorded as exploratory directions for later phases rather than committed components, in keeping with the build-order discipline.

References

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Note on citations. Bibliographic details (volumes, issues, pages, DOIs) were verified against publisher and indexing records (Taylor & Francis, Oxford Academic, SSRN/NBER) at the time of writing. The reported performance figures are those stated by Avellaneda and Lee (2010) for their sample periods and should be cited as such, not as guarantees of future or out-of-sample performance.